



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to VEESUMANI TRANSPORTS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

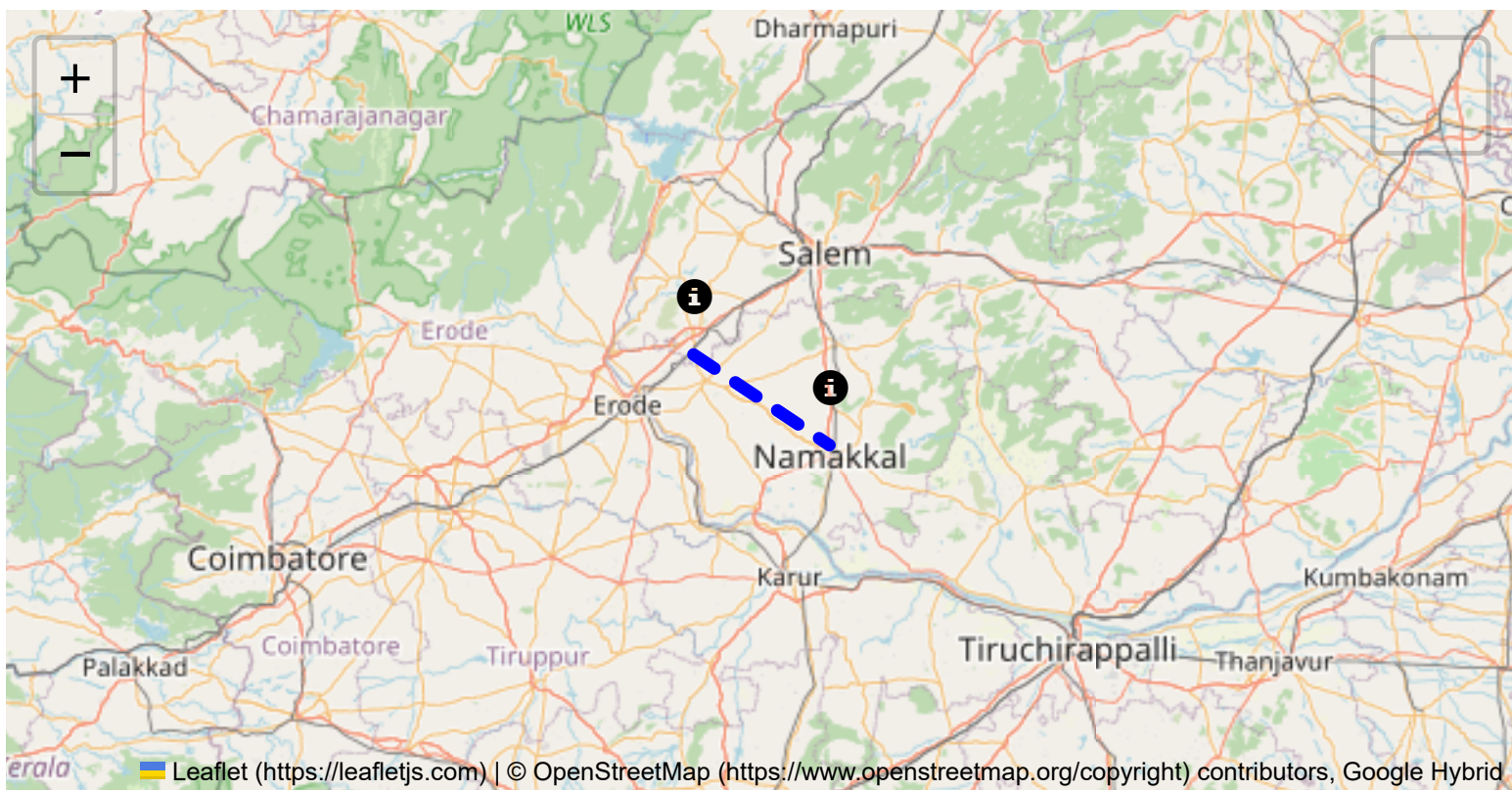
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

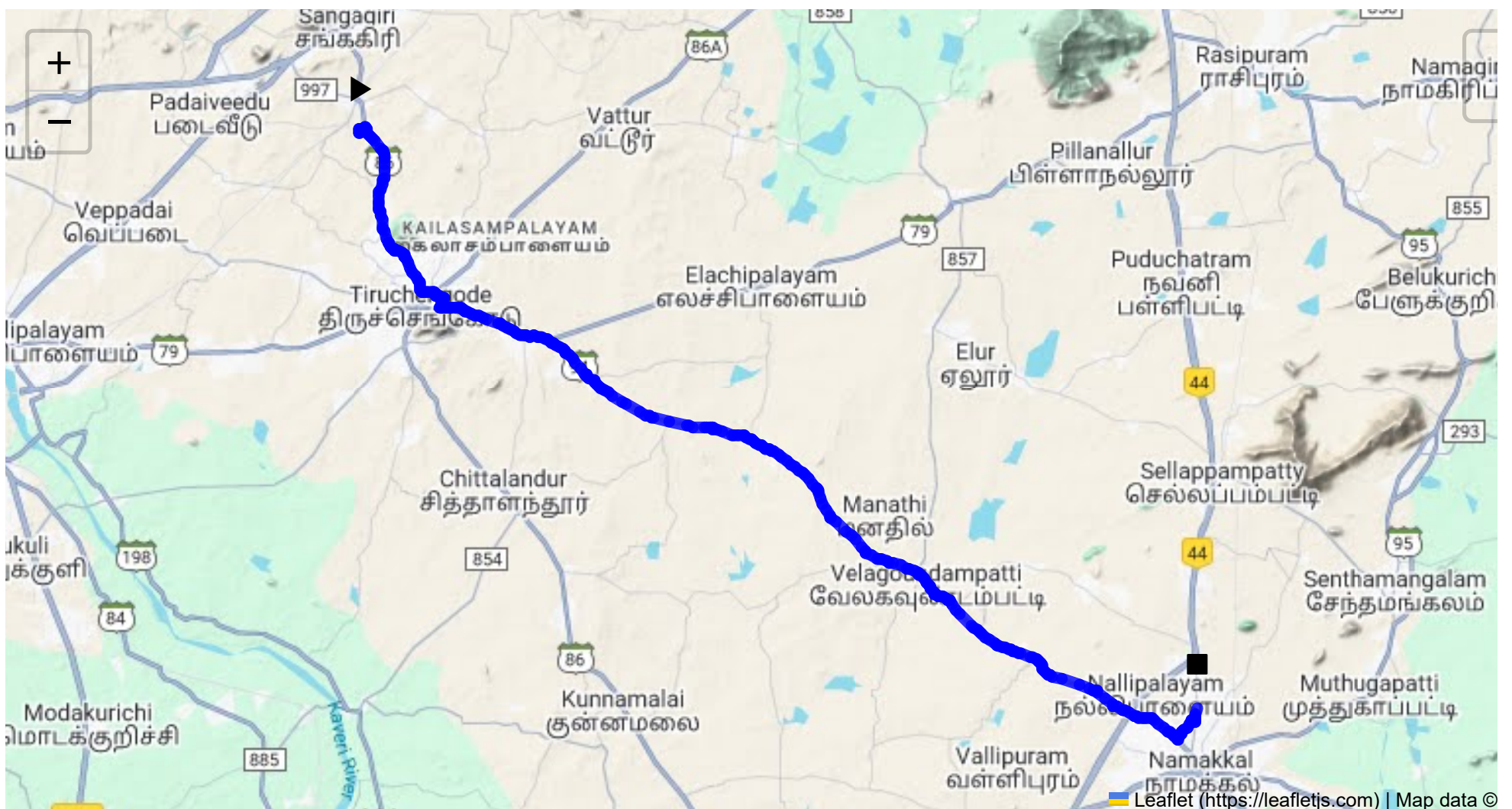
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.20 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.240785, 78.165959)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 130a, Bommasamudram via Salem-Namakkal Road covers approximately 44.20 kilometers. The journey typically takes around 1 hour for heavy vehicles carrying hazardous materials. This route navigates a mix of rural and semi-urban landscapes primarily along well-known highways.

Typical Weather Conditions

Tamil Nadu generally experiences tropical weather. Common conditions along the route include:

- **Summer (March to June):** High temperatures which may lead to overheated engines.
- **Monsoon (July to September):** Heavy rains leading to waterlogging and reduced visibility.
- **Winter (October to February):** Mild temperatures, generally more conducive to driving.

Weather-Related Hazards: During monsoon, flooding can occur, especially in low-lying areas and near water bodies, posing a risk of hydroplaning and increased accident probability.

Traffic Patterns

Traffic patterns indicate:

- **Morning Rush:** Increased congestion from 8:00 AM to 10:00 AM.
- **Evening Rush:** Heavy traffic from 5:00 PM to 7:00 PM, particularly near Salem where urban sprawl contributes to congestion.

Areas around Salem are particularly prone to bottlenecks due to cross-city traffic and daily commuters.

Road Quality and Infrastructure

The roads on this route are generally in good condition, with the major highways being well-maintained.

- **Potholes and Patches:** These may be present in rural stretches, more likely after monsoon.
- **Signage:** Adequate but keep an eye out for local detour signs and notices, particularly during roadworks.

Emergency Alternate Routes

In emergencies, the alternative route through the Erode bypass could be viable, albeit longer. Utilize navigational aids or local guidance for on-ground detours when necessary.

Local Regulations on Hazardous Material Transport

Tamil Nadu enforcement strictly regulates hazardous material transport:

- **Transportation Permits:** Required for carrying hazardous goods.
- **Restricted Hours:** Often from 6 AM to 9 PM in populated urban areas to avoid peak hours.

Historical Incidents

While the route has not been heavily publicized for major incidents recently, heavy vehicle accidents are not uncommon on major highways in the region, often attributed to driver fatigue and weather conditions.

Environmental Considerations

The route passes through semi-rural areas with agricultural activities. Minimize pollution and waste disposal to prevent environmental damage. No significant wildlife crossings reported, but vigilance is advised.

Communication Coverage

The road generally falls under strong network coverage circles, including major telecom operators. However, certain rural areas may experience patchy coverage – advisable to carry emergency communication devices.

Emergency Response Times

- **Urban Areas near Salem:** Typically quick response due to proximity to emergency services.
- **Rural Stretches near Sangagiri/Namakkal:** Response times might be longer, averaging 20-30 minutes due to distance from emergency facilities and slower speeds on narrower roads.

Overall Summary of Risk Assessment

The route presents moderate risk primarily due to weather and traffic congestion around Salem. The good quality of roads mitigates accident risk significantly. Planning travel during non-peak hours and ensuring robust communication means will contribute to a smoother transit for hazardous material transport.

Consistent updating with the latest traffic and weather reports, along with familiarity with emergency contacts is crucial for mitigating risks effectively along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.23044, 78.15943	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	High	11.23678, 78.16438	15 KM/Hr
11	Blind Spot	Blind Spot	11.23661, 78.16558	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
16	hospital	Shree Akhshaya Hospital	11.2356716, 78.165237	30 km/h	Medium
17	hospital	Bharathi Hospital	11.2342127, 78.1659819	30 km/h	Medium
18	hospital	Alpha Life Urology & Andrology Centre	11.2394378, 78.165596	30 km/h	Medium
19	hospital	B.M Hospital	11.242188, 78.166518	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
15	school	Maruthi Vidyalaya School	11.236746, 78.160895	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258

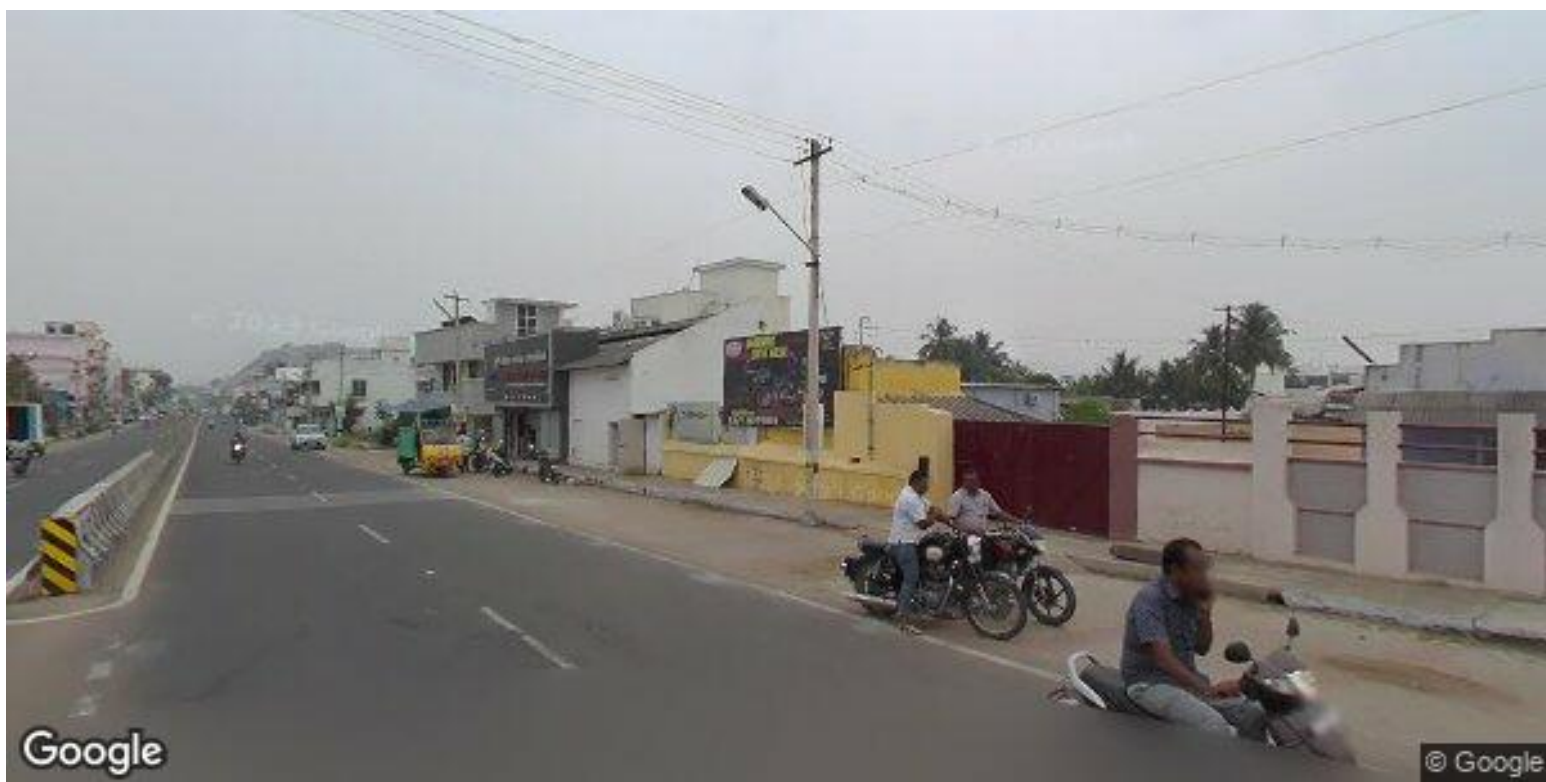


Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23044, 78.15943



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.23678, 78.16438



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23661, 78.16558
